# The Markovian Order Book

From zero intelligence to Cont-Stoikov-Talreja: model, estimation, evidence

Yaohui Shu

Stony Brook University

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# The question

Can order flow with **no strategy at all** explain the shape of the book, the spread, and how fast prices move?

- If yes, the book's statics are set by a few rates, and much of what looks like behaviour is mechanics.
- If no, the gap tells us where strategy and information matter.

For a liquidity provider (LP) this model is the **null hypothesis**: fill probabilities, queue depletion and spread you should expect before any alpha or toxicity.

## In this deck

1. The model and what it predicts exactly.
2. Estimating its rates from a message stream, with standard errors.
3. What real books show.
4. Where the model fails.

# The model

Prices on a grid with tick $\delta$ (currency). State: $X_p(t)$, the signed number of resting unit orders at price $p$ (ticks), positive for asks. Best bid $p_B$, best ask $p_A$, spread $s = p_A - p_B$ (ticks). For an order at distance $i \geq 1$ ticks from the *opposite* best quote:

| Event | Rate | Unit |
|---|---|---|
| limit buy at $p_A - i$ (sell at $p_B + i$) | $\lambda(i)$ | orders $\mathrm{s}^{-1}$ |
| market buy (sell), takes one order at $p_A$ ($p_B$) | $\mu$ | orders $\mathrm{s}^{-1}$ |
| cancellation of one resting order at distance $i$ | $\theta(i)$ per order | $\mathrm{s}^{-1}$ |

- **Zero intelligence** (Smith et al., 2003): $\lambda$ and $\theta$ constant over a window of $L$ ticks.
- **CST** (Cont et al., 2010): $\lambda(i) = \lambda_1 i^{-a}$, with $\lambda_1$ ($\mathrm{s}^{-1}$) the rate at $i = 1$ and $a$ a dimensionless exponent, fitted to Tokyo Stock Exchange data.
- All events are independent Poisson streams given the state, so $X$ is a continuous-time Markov chain. It is simulated exactly by the Gillespie algorithm.

# What the model predicts

**Depth far from the quotes.** A level that market orders rarely reach is an $M/M/\infty$ queue: arrivals at rate $\lambda(i)$, each order leaves at rate $\theta(i)$. Its mean depth is

$$\bar{Q}(i) \to \lambda(i)/\theta(i) \quad \text{(orders)}.$$

Near the quotes, market orders thin the book, so $\bar{Q}(i) < \lambda(i)/\theta(i)$.

**Dimensional analysis** (Smith et al., 2003). With prices in ticks and sizes in orders, every stationary statistic depends only on

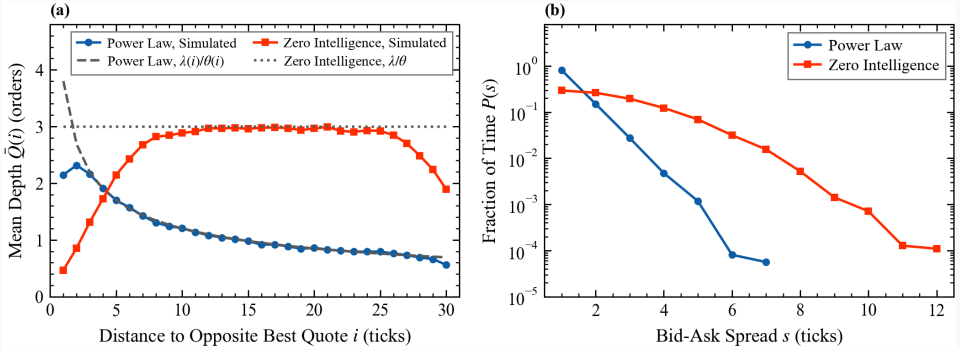
$$\lambda/\theta \text{ (depth scale)}, \quad \mu/\lambda \text{ (price scale, ticks)}, \quad L.$$

**Conditional probabilities.** CST compute, by Laplace transforms of first-passage times of birth-death queues, the probability that the mid moves up, that an order at the bid executes before the ask moves, and that a limit order fills before the price moves.

These are the same first-passage problems as in the best-queue model (deck 2), with state-dependent depletion $\mu + \theta q$.

**Assumptions.** Unit order size; rates depend on distance only, not on queue sizes or history; no information.

# Simulated depth profile and spread




- Rates: power law $\lambda_1 = 1.9\ \text{s}^{-1}$, $a = 0.5$, $\theta = 0.5\ \text{s}^{-1}$, $\mu = 1\ \text{s}^{-1}$; zero intelligence $\lambda = 0.6$, $\theta = 0.2$, $\mu = 2\ \text{s}^{-1}$. 300,000 events each.
- Mean spread: $1.226 \pm 0.006$ ticks (power law), $2.610 \pm 0.028$ ticks (zero intelligence). Far depth 0.56 against the M/M/$\infty$ value 0.69 orders.

# Estimation: the likelihood of a Markov jump process

Observe the book on $[0, T]$ (s). For a Markov jump process the log-likelihood is

$$\ell = \sum_{\text{events } k} \ln r_{e_k}(X_{t_k^-}) \; - \; \int_0^T \sum_e r_e(X_t)\,\mathrm{d}t,$$

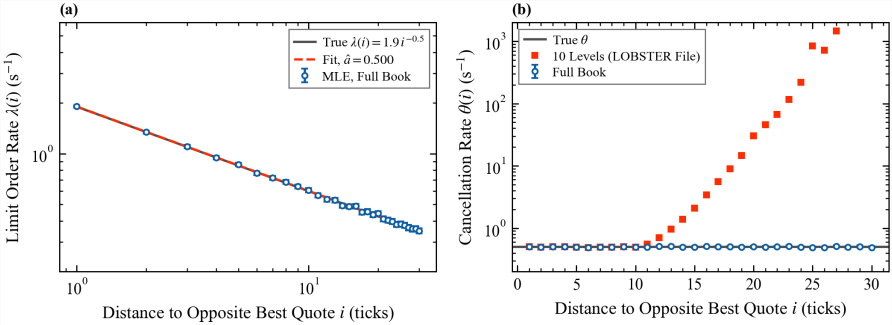
with $r_e(x)$ the rate of event type $e$ in state $x$. Because each rate depends on one parameter, $\ell$ splits into independent Poisson problems:

$$\hat{\lambda}(i) = \frac{N_L(i)}{2T}, \qquad \hat{\theta}(i) = \frac{N_C(i)}{E(i)}, \qquad \hat{\mu} = \frac{N_M}{2T},$$

where $N_L(i)$, $N_C(i)$, $N_M$ count limit orders and cancellations at distance $i$ and market orders (both sides), and $E(i) = \int_0^T [\text{orders at distance } i]\,\mathrm{d}t$ (order-seconds) is the **exposure**. Standard errors: $\sqrt{N}$/denominator.

- Distances are measured against the quotes *just before* each event.
- Cancellation rates need the exposure, so they need the full depth over time, not just the messages.

# Estimation in practice: recovery, and the depth-truncation trap




300,000 events (1.1 h) of the power-law book, written to and read back from LOBSTER files. (a) $\hat{\lambda}(i)$ within 2.5 s.e. of the truth; weighted least squares on $\ln \hat{\lambda}$ gives $\hat{a} = 0.500 \pm 0.002$ (true 0.5). $\hat{\mu} = 1.005 \pm 0.011$ s$^{-1}$.

(b) With the full book, $\hat{\theta}(i)$ is within 1.6 s.e. A 10-level file sees about 13.1 ticks of book, so the exposure is cut off while cancellation messages still arrive: $\hat{\theta}(i)$ is biased from $i = 11$ on.

# What real books show

- **Placement is broad.** On the Paris Bourse (France Telecom, Vivendi, Total; February 2001), the distance of new limit orders from the current price follows a power law with exponent about 0.6, from 1 to over 100 ticks. The mean book is **hump-shaped**: depth peaks away from the best quotes (Bouchaud et al., 2002).
- **Zero intelligence explains cross-sectional statics.** For 11 London Stock Exchange stocks, the model with measured rates explains 96% of the variance of the mean spread and 76% of the variance of the price diffusion rate across stocks (Farmer et al., 2005).
- **Power-law placement fits.** CST estimate a power-law $\lambda(i)$ from Tokyo Stock Exchange order books and use the fitted model to compute conditional probabilities of price moves (Cont et al., 2010).
- **Survey.** Gould et al. (2013) collect the stylized facts and compare models against them.

Numbers above are as reported in the cited papers; this repository uses no market data.

# Where the model fails

**State dependence.** Real intensities depend on queue size: market orders arrive faster when the best queue is short, and cancellations per order fall as queues grow (Huang et al., 2015). Deck 3.

**Memory.** Order signs are long-range correlated, with autocorrelation decaying as $\ell^{-0.6}$ on the LSE (Lillo and Farmer, 2004); here they are independent. Deck 5.

**Self-excitation.** Events cluster in time beyond what the state explains. Deck 4.

**Information.** Nobody knows anything, so there is no adverse selection from information. Yet even here, the fills that empty a queue are followed by an adverse move: they keep $0.04 \pm 0.01$ ticks against $0.61 \pm 0.00$ for other fills (deck 8). Queue mechanics alone create toxicity.

**Order size.** Unit orders; real orders vary over orders of magnitude.

# Takeaways for a liquidity provider

1. **Use it as a baseline.** Fill probabilities, depletion times and spread from a model fitted to today's rates are the benchmark that a strategy's edge is measured against.
2. **Estimate with exposure, not counts.** Cancellation rates per order need time-integrated depth. A rate computed as "cancels per second" confuses a busy level with a deep one.
3. **Mind the data depth.** Ten visible levels cover about 13.1 ticks here; beyond that, cancellation estimates are wrong, and so is anything built on them.
4. **Standard errors are cheap.** The Poisson likelihood gives them for free; report them.

**Open questions.** How much of the cross-section of spreads and volatility in today's fragmented, high-frequency markets do rate-only models still explain? Which state variables (queue sizes, recent flow, time of day) should enter the rates first?

# Reproduce

Code: `src/stochastic_lob/markov_lob.py` (simulation, `estimate_rates`), `lobster.py` (data format). Experiments: `e1_markov_book.py`, `e15_rate_estimation.py`.

```
from stochastic_lob import markov_lob as ml
p = ml.power_law_params(L=30, lam1=1.9, a=0.5, theta0=0.5, mu=1.0)
rec = ml.simulate_markov_lob(p, 200_000, rng=1, record_levels=40)
est = ml.estimate_rates(rec, L=30)     # lam, theta, mu with standard errors
```

With LOBSTER files: `lobster.to_record(lobster.read(msg, book))` gives the same record.

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